

Employee Management System

Java Program Report

1. Introduction

This is a console-based Employee Management System written in Java. It allows the user to create an employee record, display employee details, and increase the employee salary. The program demonstrates classes, objects, methods, loops, conditional statements, Scanner input, and validation.

2. Objectives

- Create and store an employee record.
- Validate the employee name and age.
- Select Programmer, Manager, or Tester as designation.
- Display employee information and salary.
- Increase the employee salary.

3. Description of the Program

Employee class: Stores name, age, designation, and salary. The default salary is ■99,999.

create(): Accepts name, age, and designation. It validates the name, restricts age to 16–66, and lets the user edit the details.

display(): Shows the saved employee details and salary.

raiseSalary(): Reads a salary increase, adds it to the current salary, and displays the updated amount.

main(): Shows the menu and repeatedly executes the selected operation until the user chooses Exit.

4. Program Flow

Start → Menu → Create / Display / Raise Salary → Perform Operation → Return to Menu → Exit.

5. Code Screenshot

BankingSystemCode2.java

```
01 package BankingSystemProgram;
02
03 import java.util.Scanner;
04
05 class Employee {
06     String name;
07     int age;
08     String designation;
09     double salary = 99999;
10 }
11
12 public class BankingSystemCode2 {
13     static Scanner sc = new Scanner(System.in);
14     static Employee e = new Employee();
15
16     static void create() {
17         while (true) {
18             while (true) {
19                 System.out.print("Enter your name: ");
20                 e.name = sc.nextLine();
21                 int spaces = 0;
22                 for (int i = 0; i < e.name.length(); i++) {
23                     if (e.name.charAt(i) == ' ')
24                         spaces++;
25                 }
26                 if (e.name.isEmpty())
27                     System.out.println("Name cannot be empty!");
28                 else if (spaces > 2)
29                     System.out.println("Maximum 2 spaces allowed!");
30                 else
31                     break;
32             }
33
34             while (true) {
35                 System.out.print("Enter your age: ");
36                 e.age = sc.nextInt();
37                 sc.nextLine();
38                 if (e.age >= 16 && e.age <= 66)
39                     break;
40                 System.out.println("Age must be between 16 and 66!");
41             }
42
43             while (true) {
44                 System.out.println("\n1. Programmer");
45                 System.out.println("2. Manager");
46                 System.out.println("3. Tester");
47                 System.out.print("Select designation: ");
48                 int d = sc.nextInt();
49                 sc.nextLine();
50
51                 if (d == 1) {
52                     e.designation = "Programmer";
53                     break;
54                 } else if (d == 2) {
55                     e.designation = "Manager";
56                     break;
57                 } else if (d == 3) {
58                     e.designation = "Tester";
```

6. Sample Result and Output

The following sample run shows creating an employee, displaying the employee record, raising the salary by ₹5,000, and exiting the program.

```
===== Employee Management =====
1. Create
2. Display
3. Raise Salary
4. Exit
Enter choice: 1

Enter your name: Ujwal Sherikar
Enter your age: 21

1. Programmer
2. Manager
3. Tester
Select designation: 1

--- Employee Details ---
Name: Ujwal Sherikar
Age: 21
Designation: Programmer
Do you want to edit? (yes/no): no
Employee saved!

===== Employee Management =====
1. Create
2. Display
3. Raise Salary
4. Exit
Enter choice: 2

--- Employee Details ---
Name: Ujwal Sherikar
Age: 21
Designation: Programmer
Salary: ₹99999.0

===== Employee Management =====
1. Create
2. Display
3. Raise Salary
4. Exit
Enter choice: 3

Current Salary: ₹99999.0
Enter salary raise: ₹5000
Salary updated!
New Salary: ₹104999.0

===== Employee Management =====
1. Create
2. Display
3. Raise Salary
4. Exit
Enter choice: 4
Thank you for using Employee Management System!
```

Result: The employee record was created successfully, the details were displayed correctly, and the salary was updated from ₹99,999 to ₹1,04,999 after adding a ₹5,000 raise.

7. Complete Source Code

```
package BankingSystemProgram;

import java.util.Scanner;

class Employee {
    String name;
    int age;
    String designation;
    double salary = 99999;
}

public class BankingSystemCode2 {
    static Scanner sc = new Scanner(System.in);
    static Employee e = new Employee();

    static void create() {
        while (true) {
            while (true) {
                System.out.print("Enter your name: ");
                e.name = sc.nextLine();
                int spaces = 0;
                for (int i = 0; i < e.name.length(); i++) {
                    if (e.name.charAt(i) == ' ')
                        spaces++;
                }
                if (e.name.isEmpty())
                    System.out.println("Name cannot be empty!");
                else if (spaces > 2)
                    System.out.println("Maximum 2 spaces allowed!");
                else
                    break;
            }

            while (true) {
                System.out.print("Enter your age: ");
                e.age = sc.nextInt();
                sc.nextLine();
                if (e.age >= 16 && e.age <= 66)
                    break;
                System.out.println("Age must be between 16 and 66!");
            }

            while (true) {
                System.out.println("\n1. Programmer");
                System.out.println("2. Manager");
                System.out.println("3. Tester");
                System.out.print("Select designation: ");
                int d = sc.nextInt();
                sc.nextLine();

                if (d == 1) {
                    e.designation = "Programmer";
                    break;
                } else if (d == 2) {
                    e.designation = "Manager";
                    break;
                } else if (d == 3) {
                    e.designation = "Tester";
                    break;
                } else {
                    System.out.println("Invalid choice!");
                }
            }

            System.out.println("\n--- Employee Details ---");
            System.out.println("Name: " + e.name);
            System.out.println("Age: " + e.age);
            System.out.println("Designation: " + e.designation);

            System.out.print("Do you want to edit? (yes/no): ");
            String answer = sc.nextLine();
            if (answer.equalsIgnoreCase("no")) {
                System.out.println("Employee saved!");
                break;
            }
        }
    }

    static void display() {
        if (e.name == null) {
            System.out.println("No employee records available.");
        } else {
            System.out.println("\n--- Employee Details ---");
            System.out.println("Name: " + e.name);
            System.out.println("Age: " + e.age);
            System.out.println("Designation: " + e.designation);
            System.out.println("Salary: ■" + e.salary);
        }
    }

    static void raiseSalary() {
        if (e.name == null) {
            System.out.println("No employee records available.");
        }
    }
}
```

```

        return;
    }
    System.out.println("Current Salary: ■" + e.salary);
    System.out.print("Enter salary raise: ■");
    double raise = sc.nextDouble();
    sc.nextLine();
    e.salary = e.salary + raise;
    System.out.println("Salary updated!");
    System.out.println("New Salary: ■" + e.salary);
}

public static void main(String[] args) {
    int choice;
    do {
        System.out.println("\n===== Employee Management =====");
        System.out.println("1. Create");
        System.out.println("2. Display");
        System.out.println("3. Raise Salary");
        System.out.println("4. Exit");
        System.out.print("Enter choice: ");

        choice = sc.nextInt();
        sc.nextLine();

        if (choice == 1)
            create();
        else if (choice == 2)
            display();
        else if (choice == 3)
            raiseSalary();
        else if (choice == 4)
            System.out.println(
                "Thank you for using Employee Management System!"
            );
        else
            System.out.println("Invalid choice!");
    } while (choice != 4);
}
}

```

8. Validation

• Name cannot be empty. • Maximum two spaces are allowed. • Age must be between 16 and 66. • Designation must be 1, 2, or 3. • Display and salary raise require an existing employee record.

9. Conclusion

The program is a simple example of Java employee management using object-oriented programming and basic input validation. The sample execution confirms that the main operations work as intended.